

Exploratory Data Analysis On Climate Concerns Of Bangladesh

Kazi Sadat Hossain
kazisadath@gmail.com

Abstract—Bangladesh’s climate has gone through a rigorous and consistent change. A dataset has provided a long-term perspective on how this has conspired by the last 20 years. By examining deviations in temperature, heat stress indicators, rainfall distribution, sunshine duration, and wind behavior and emerging anomalies within the country’s climate system. This approach makes it easier to spot extreme weather events, changes in monsoon timing, increasing heat-stress days, and shifts in storm intensity. By breaking the climate data down season by season, we can see what’s normal and what’s starting to change. Altogether, these insights give an early warning of how Bangladesh’s climate may be shifting, helping guide smarter decisions in farming, infrastructure, public health, and environmental planning.

Keywords—climate analysis, exploratory data analysis, temperature deviation, monsoon patterns, heat stress, extreme weather events, Bangladesh climate, seasonal profiling.

I. INTRODUCTION

Climate change has emerged as one of the most pressing global challenges of the 21st century, and its impacts are particularly evident in countries like Bangladesh. Situated in a low-lying delta region, Bangladesh is highly vulnerable to the consequences of changing weather patterns, rising temperatures, and shifting rainfall trends. Over the past few decades, observational data—including the daily weather records from 1995 onwards—provides clear evidence of these climatic changes. One of the most striking trends is the steady increase in temperature. Analysis of 5-year aggregated data shows that the average mean temperature in Bangladesh has risen from around 17.5°C in the late 1990s to nearly 20°C in recent years, while maximum temperatures have also increased by approximately 2–3°C over the same period. Summers are becoming hotter, and even nights are warmer than before. This rising heat affects everyone—from farmers struggling to grow crops under scorching conditions, to city residents facing heat stress. Higher temperatures also worsen other challenges, such as water scarcity, and make extreme weather events even more dangerous.

Equally concerning are the changes in rainfall patterns. The dataset reveals that some 5-year periods have experienced heavier cumulative rainfall, leading to flooding in low-lying areas, while other periods saw lower-than-average rainfall, contributing to drought conditions. For example, total rainfall over one 5-year period exceeded 10,000 mm, while another period recorded only around 8,500 mm, showing significant variability. These unpredictable patterns disrupt agriculture, as farmers cannot rely on consistent rains for planting and harvesting. Crops fail, food becomes scarce, and livelihoods

are threatened. Heavy rainfall also leads to urban flooding, waterlogging, and riverbank erosion, directly impacting communities near rivers.

The combination of rising heat and fluctuating rainfall paints a challenging picture. It is not just numbers on a chart—these changes are felt in everyday life. Children are at higher risk during heatwaves, farmers face uncertainty each season, and entire communities must cope with floods and water shortages. The historical weather data helps us understand these patterns, showing that climate change in Bangladesh is gradual but very real, with tangible impacts on people, agriculture, and ecosystems.

In conclusion, Bangladesh’s experience with climate change is a clear warning about the consequences of rising temperatures and unpredictable rainfall. By analyzing weather data, we can see that average temperatures have steadily increased and rainfall has become increasingly erratic over the last three decades. This knowledge is essential for planning better adaptation strategies, strengthening disaster preparedness, and protecting the communities most vulnerable to a changing climate. Every rising degree and every unpredictable downpour tells a story—and for Bangladesh, it is a story that must be addressed now, before the impacts become even more severe.

II. DATASET

A. Selecting a Dataset

A dataset was selected from kaggle that most suited the exploratory data analysis subject. The data given there was a global accumulation of weather report from 1995-2024. The dataset was later filtered to use only the data concerning Bangladesh. Thus providing a deep insight into the change of weather for the past 25 years and giving a rough estimation of climate change that is happening.

B. Dataset variables

Overall the dataset has 20 columns and 10958 rows after filtering. The filtered data covers a variety of variables that is required to thoroughly analyze the dataset.

TABLE 1. DATASET ATTRIBUTES

Attribute description	
date	Specific date of the observation(Continuous)
Country	Country of the observation(Nominal)
country_alpha 2	Two-letter ISO code for the country (Nominal)
Capital	Capital city (Nominal)
Lat	Latitude of location(Continuous)
Lon	Longitude of location(Continuous)
temp_min_c	Minimum daily temperature in °C(Continuous)
temp_max_c	Maximum daily temperature in °C(Continuous)
temp_mean_c _approx	Approximate mean daily temperature in°C(Continuous)
app_temp_min _c	Minimum apparent temperature in°C(Continuous)
app_temp_ma x_c	Maximum apparent temperature in°C(Continuous)
precip_mm	Total daily precipitation in mm(Continuous)
rain_mm	rainfall in mm(Continuous)
snow_mm	Snowfall in mm(Continuous)
windspeed_10 m_max_kmh	Maximum wind speed in km/h (Continuous)
windgusts_10 m_max_kmh	Maximum wind gusts in km/h (Continuous)
wind_dir_dom _deg	Dominant wind direction in degrees (Ordinal/Continuous)
sunshine_durat ion_s	Total sunshine duration in seconds (Continuous)
daylight_durat ion_s	Total daylight duration in seconds (Continuous)
shortwave_rad iation_MJ_m2	Incoming solar radiation in MJ/m²(Continuous)

III. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is a crucial first step in understanding and interpreting any dataset. It allows us to uncover underlying patterns, detect anomalies, test hypotheses, and extract meaningful insights before performing more advanced modeling or statistical analysis. In this study, we focus on the dataset, which contains daily weather observations spanning several decades, including temperature, rainfall, wind speed, sunshine duration, and other meteorological variables.

The objective of this EDA is to provide a comprehensive overview of climate trends and variability in Bangladesh. By analyzing data over time, particularly in 5-year aggregated periods, we aim to identify changes in average temperature, precipitation, wind patterns, and sunlight duration. Visualizations such as line plots, bar charts, correlation heatmaps, and pie charts are employed to make patterns clear and interpretable.

Through this analysis, we can better understand temporal trends in weather and climate, observe seasonal variations, and detect potential indicators of climate change, providing a solid foundation for future environmental and meteorological studies.

The data analysis was done thoroughly and various segments are given below.

1) *Rainfall*

The piechart below is a pie chart titled “Annual Maximum Temperature Distribution Across Years (Bangladesh)”, showing how the highest recorded temperatures from 1995 to 2024 contribute proportionally to the overall dataset. Each year is represented by a thin, colored slice with its percentage labeled. The values stay very close to each other—mostly between 3.1% and 3.7%—indicating that maximum temperatures across these years do not vary dramatically. The chart appears visually balanced and uniform, suggesting that although temperatures may fluctuate slightly year to year, the overall distribution remains relatively stable without extreme spikes.

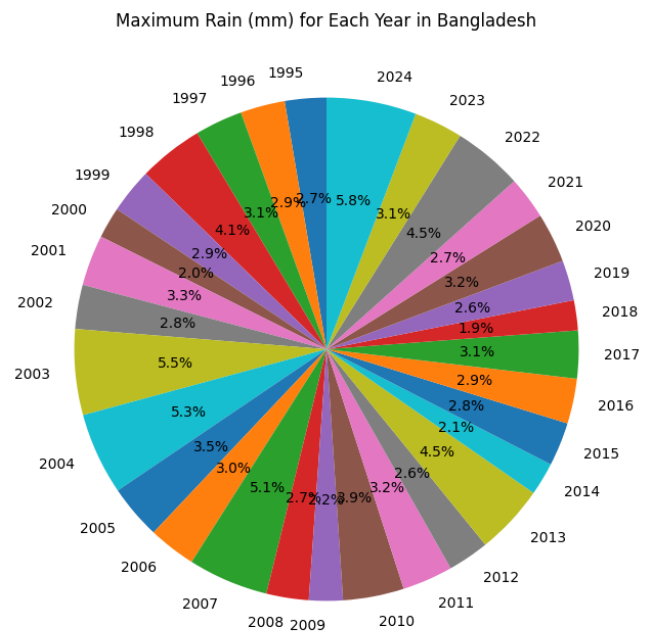


Fig 01: Rainfall distribution of each year

2) *Heat*

The piechart below is a pie chart titled “Maximum Rain (mm) for Each Year in Bangladesh)”, illustrating the proportional distribution of the highest annual rainfall amounts from 1995 to 2024. Unlike the temperature chart, the slice sizes vary more noticeably. Some years show relatively small contributions (around 1.9% to 2.9%), while others stand out with significantly larger percentages (around 5% to 5.8%). This reveals that maximum rainfall levels fluctuate more widely from year to year, with certain years experiencing unusually high peak rainfall. Visually, the chart appears more uneven compared to the temperature chart, reflecting the greater variability in extreme rainfall events over the years.

Annual Maximum Temperature Distribution Across Years (Bangladesh)

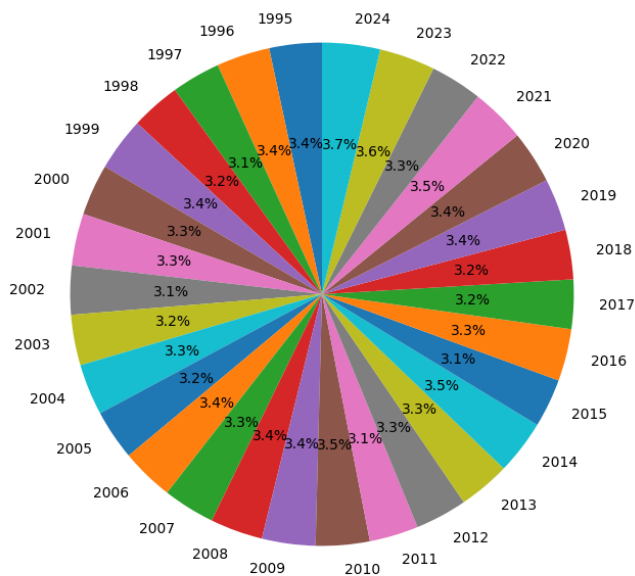


Fig 02: Heat distribution of each year

3) Mean Temperature Trend

The image is a time-series line graph titled “Daily Mean Temperature Trend — Bangladesh”, showing fluctuations in average daily temperature from around 1995 to 2024. The temperature values on the y-axis range roughly from 13°C to 35°C, and the pattern forms a clear repeating cycle each year. These strong, regular oscillations represent the seasonal temperature changes in Bangladesh, with warmer peaks in the summer months and cooler dips during winter. Although the year-to-year variations follow a consistent seasonal rhythm, the upper peaks appear to gradually rise in the later years—especially after 2015—suggesting a subtle warming trend over the long term. Overall, the chart captures nearly three decades of daily climate behavior and visually highlights both the predictability of seasonal changes and the broader trend of increasing temperatures.

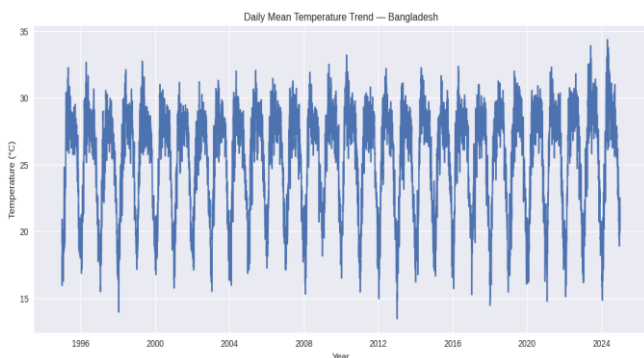


Fig 03: Mean Temperature Trend in C

4) Mean Temperature Distribution

This image is a histogram titled “Bangladesh – Mean Temperature Distribution”, illustrating how daily mean temperatures are distributed over a long period. The x-axis represents temperature values (in °C), ranging from around 14°C to 35°C, while the y-axis shows the count of days that fall within each temperature range. The histogram bars

display the frequency of temperatures, and a smooth density curve overlays the bars to highlight the overall shape of the distribution.

The graph shows that most daily mean temperatures cluster between 26°C and 31°C, with a clear peak around 28–30°C, indicating that warm temperatures are very common in Bangladesh. There is a noticeable smaller hump around 20–24°C, representing cooler periods, likely winter months. The distribution is moderately skewed, with fewer extremely low or high temperatures. Overall, the chart reveals that Bangladesh predominantly experiences warm daily mean temperatures, with a secondary cluster of cooler days and relatively few occurrences of very low or very high temperature values.

There is another figure provided to further explains the monthly mean temperature and the accumulation that has conspired the initial graph.

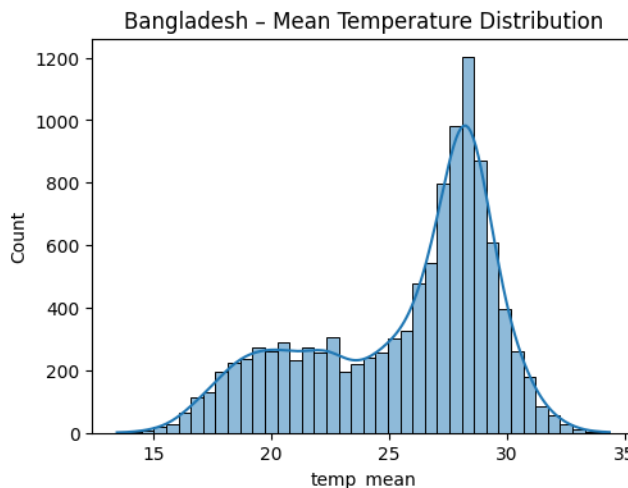


Fig 04:Mean temperature distribution

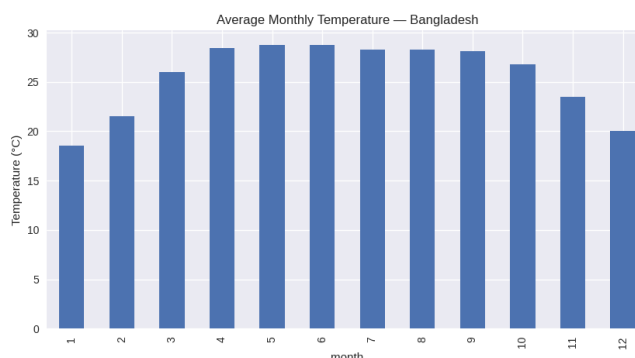


Fig 05: Monthly Mean Temperature Distribution

5) Annual Mean Temperature

This image is a line chart titled “Bangladesh – Annual Mean Temperature”, showing the average yearly temperature from 1995 to 2024. Each point represents the mean temperature for that year, connected by a line to reveal the overall trend. The

values fluctuate between roughly 25.0°C and 26.4°C, indicating that annual mean temperatures vary within a relatively narrow range.

Despite these yearly ups and downs, the general pattern suggests a gradual warming trend over the nearly three-decade period. Temperatures in the late 1990s and early 2000s mostly hover around 25.0–25.5°C, while more recent years, especially after 2010, tend to reach higher averages closer to 25.8–26.2°C. Notable peaks occur around 2007, 2010, 2016, and 2023, with 2010 showing the highest mean temperature in the dataset. The chart highlights that Bangladesh has experienced slow but steady warming over time, consistent with broader climate change patterns, although year-to-year variability remains significant

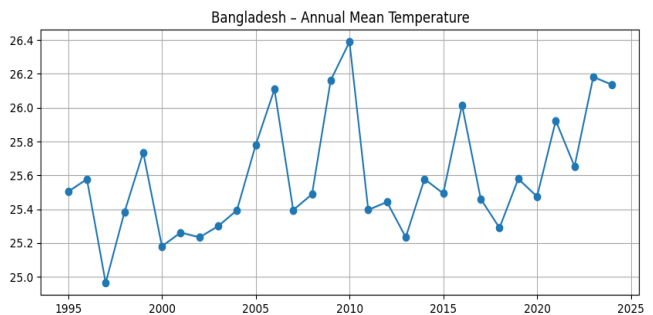


Fig 06: Annual Mean Temperature

6) Temperature analysis

This image is a boxplot titled “Bangladesh – Temperature Boxplot,” which summarizes the distribution of daily mean temperatures. The horizontal axis shows temperature values in degrees Celsius, while the vertical axis does not represent categories because the chart focuses solely on temperature variation.

The boxplot indicates that the middle 50 percent of temperatures (the interquartile range) falls roughly between 25°C and 30°C, showing that most days in Bangladesh remain within this warm range. The median temperature, represented by the line inside the box, is around 28°C, which reflects the country’s generally warm climate. The whiskers extend from about 15°C to 34°C, covering the coolest winter days and the hottest summer days. There is also a small outlier near the lower end, representing an unusually cool day compared to the rest of the dataset. Overall, the boxplot highlights that Bangladesh experiences predominantly warm temperatures with a moderate range and very few extreme outliers.

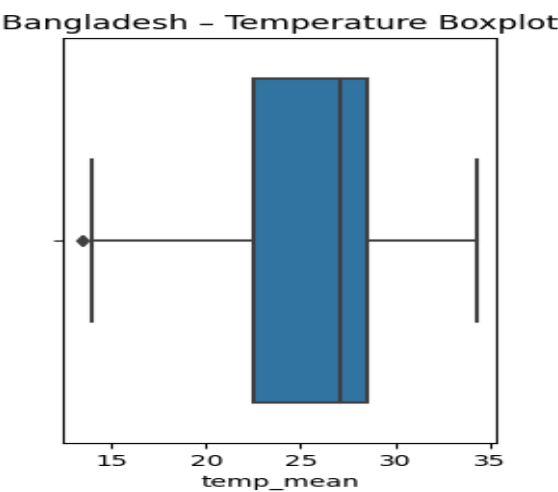


Fig 07: Mean Temperature Boxplot

7) Heat Stress Analysis

The first chart, titled “Heat-Stress Days per Year (>32°C),” is a bar graph showing the number of days each year when temperatures exceeded 32°C. The graph spans from 1995 to 2024 and reveals strong year-to-year variability. Some years in the late 1990s and early 2000s show relatively high counts, with peaks around 1995, 1998, and 2009. However, the later years show a clear upward trend, especially after 2015. The most dramatic increase appears in the early 2020s, where the number of heat-stress days rises sharply, with 2023 and 2024 showing the highest values—reaching around 40 to nearly 50 days. This pattern suggests that extreme heat events have become more frequent in recent years, indicating a growing heat-stress burden likely linked to climate warming.

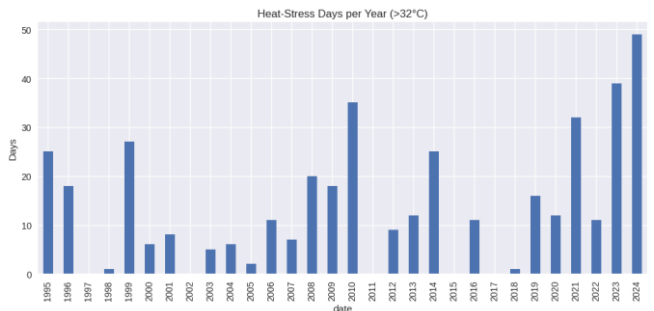


Fig 08: Heat Stress Days Per Year (>32°C)

The second chart, titled “Bangladesh – Annual Heat-Alert Days (App Temp > 32°C),” is a line graph showing how many days per year had an apparent temperature (feels-like temperature) above 32°C. The values fluctuate each year, generally ranging between about 230 and 260 days. Despite the year-to-year variations, the long-term trend appears to be gradually increasing. The early years mostly remain in the lower part of the range, while later years, especially after 2005, show more frequent peaks around 255–260 days. The highest value occurs around 2010, and several recent years also show elevated counts. This indicates that Bangladesh experiences a very high number of heat-alert days annually and that the frequency of such days has slowly increased over time, pointing toward a warming climate and intensifying heat conditions.

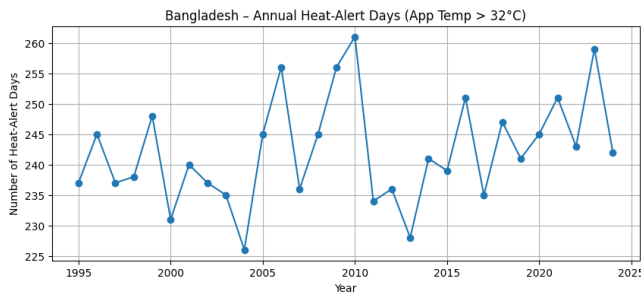


Fig 09: Annual Heat Alert Days(>32°C)

8) 5 Year Heat Average Heat trend

This chart, titled “5-Year Average Temperature Trends in Bangladesh,” presents grouped 5-year averages of mean, maximum, and minimum temperatures for several periods between 1995 and 2020. Three differently colored lines represent the three temperature categories, allowing easy comparison of how each one changes over time.

The mean temperature line stays fairly stable, fluctuating slightly between about 25.3°C and 25.9°C. There is a small dip around the year 2000, followed by modest increases in later periods, with the highest value appearing near 2020. The minimum temperature line shows more subtle movement, remaining close to 21.4°C to 21.8°C throughout the timeline. Although the changes are small, there is a gentle rising tendency during the middle years, followed by a slight decline toward 2020. The maximum temperature line remains the highest of the three and shows clearer variation. It decreases slightly around the year 2000, rises noticeably by 2005, levels out through 2010 and 2015, and then reaches its highest point near 2020.

Overall, the chart suggests that all three temperature measures have experienced mild increases over the past few decades, with the maximum temperatures showing the most noticeable upward trend. The general stability of the mean and minimum temperatures, combined with the upward shift in maximum values, points to gradually intensifying heat conditions in Bangladesh.

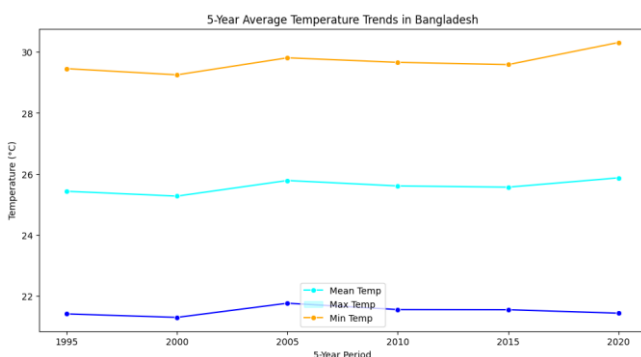


Fig 10: 5 Year average Heat Trends

9) Monthly Rain Index

The chart shows the total monthly rainfall in Bangladesh and highlights a very clear seasonal pattern. Rainfall is extremely

low during the winter and early spring months, with January and February receiving only a small amount. Starting in March, rainfall begins to rise, and by April it increases sharply. The monsoon season becomes fully active from May onward, with a dramatic surge that peaks in June at the highest rainfall level of the year. Rainfall then remains very high through July and August, though it gradually declines after June. By September and October, the rainfall continues to decrease but still stays significant compared to the dry months. Finally, in November and December, rainfall drops back to very low levels, marking the return of the dry season. The overall pattern clearly reflects Bangladesh’s monsoon-dominated climate, with intense rainfall concentrated between late spring and early autumn.

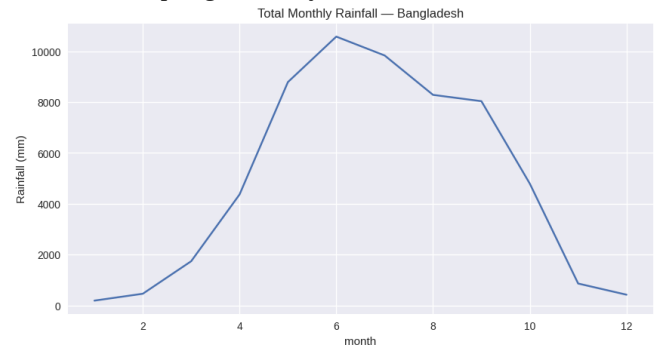


Fig 11: Monthly Rain Distribution(mm)

10) Annual rainfall analysis

The annual rainfall chart for Bangladesh shows a long-term pattern of highly variable daily rainfall across nearly three decades. Throughout the entire timeline, rainfall follows a repeating seasonal rhythm each year, characterized by long stretches of very low or near-zero rainfall during the dry months, followed by sharp spikes during the monsoon. These spikes represent heavy rainfall events, and they appear consistently every year, though their exact height varies.

Some years show particularly intense peaks, indicating episodes of extreme rainfall. These tall spikes become noticeable around the early 2000s, mid-2010s, and again in the early 2020s. Despite this variability, the overall structure does not show a clear long-term increase or decrease in annual rainfall. Instead, the pattern reflects Bangladesh’s strong monsoon cycle, where heavy rainfall clusters within a few months but remains minimal for the rest of the year. The dense concentration of spikes each year highlights how rainfall is not evenly distributed but occurs in short, high-intensity bursts typical of a monsoon-driven climate.

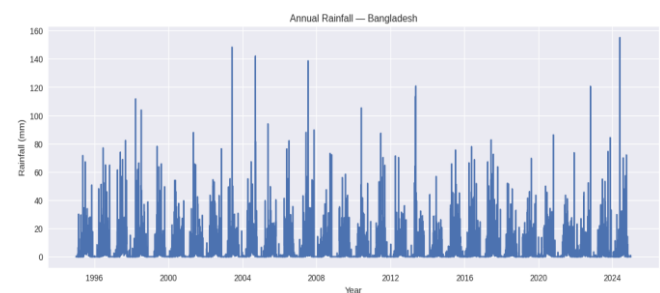


Fig 12: Annual Rainfall (mm)

This bar chart shows how Bangladesh’s total annual rainfall has fluctuated from the mid-1990s to 2024. The pattern reveals considerable year-to-year variability, with no perfectly consistent upward or downward trend, but several notable peaks and dips stand out. During the late 1990s and early 2000s, rainfall remained relatively high, with multiple years exceeding 2,200 mm. A prominent spike appears around 2007, marking one of the wettest years in the dataset.

After this peak, rainfall levels dip at times—particularly around 2013, which shows one of the lowest totals—before rising again sharply in 2017, the wettest year overall. In the more recent years, rainfall appears somewhat moderate compared to earlier extremes, fluctuating between roughly 1,600 mm and 2,000 mm, suggesting slightly reduced but still variable monsoon intensity.

Overall, the chart indicates that Bangladesh experiences substantial natural variability in monsoon rainfall, with occasional extreme years but no clear long-term direction of increase or decrease

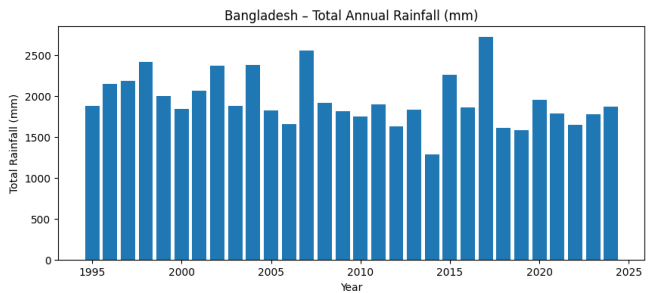


Fig 13: Total Rainfall Index(mm)

11) Heavy Rainfall Analysis

The charts show the annual number of heavy rain days in Bangladesh, defined as days with more than 50 millimeters of rainfall. Although the year-to-year values fluctuate noticeably, the overall pattern reveals recurring spikes roughly every few years. In several years such as 1997, 2004, 2007, 2011, 2016, and 2024, the number of heavy rain days rises sharply, reaching between six and ten events. Other years show much lower counts, sometimes only one or two heavy rainfall days. This irregular behavior suggests that extreme rainfall events do not follow a smooth trend but occur in bursts, possibly influenced by monsoon variability, regional storms, and large-scale climate patterns such as ENSO. Despite the fluctuations, the persistence of repeated peaks over the three decades indicates that intense rainfall remains a regular feature of Bangladesh’s climate, posing ongoing challenges for flood management and disaster preparedness.

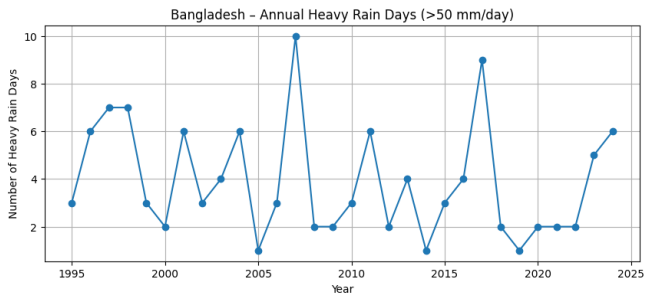


Fig 14: Annual Heavy Rain Index(>50mm)

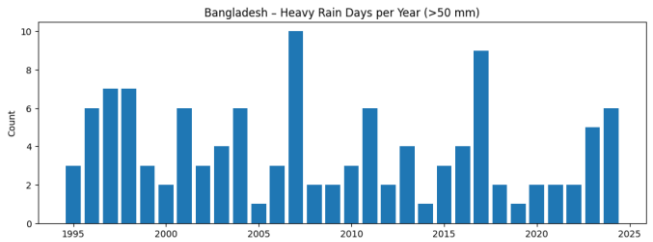


Fig 15: heavy Rain per 5 year interval Index(>50mm)

12) Rainfall Analysis of Every 5 year

The chart illustrates the total rainfall in Bangladesh over successive five-year periods, highlighting how overall precipitation has shifted across the past few decades. Rainfall remains relatively high in all periods, but there is a noticeable decline after the early years. The periods 1995 and 2000 show the highest totals, both slightly above ten thousand millimeters, indicating particularly wet intervals. After 2000, the totals begin to fall, dropping to around nine thousand eight hundred millimeters in 2005 and reaching their lowest point in 2010, where rainfall decreases to about eight thousand four hundred millimeters. A partial recovery occurs in 2015, where rainfall rises back to roughly ten thousand millimeters, before declining again in 2020 to about nine thousand millimeters. Overall, the pattern suggests that while Bangladesh continues to receive substantial rainfall, the totals vary considerably from one five-year period to the next, with signs of both short-term declines and intermittent rebounds.

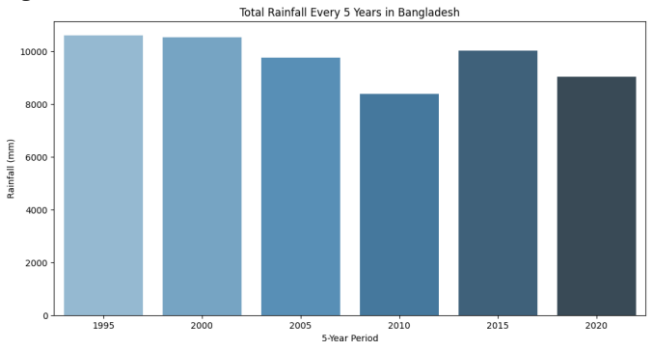


Fig 16: 5 Year Rain accumulation Index

13) Windspeed Deviation Analysis

The chart shows how Bangladesh’s average maximum wind speed has changed across six five-year intervals, revealing a pattern of decline followed by a steady rise. Wind speeds begin at just over 13.1 km/h in 1995 and gradually decrease through 2000 and 2005, reaching the lowest point at about 12.7 km/h. After this dip, the trend shifts upward: by 2010,

wind speeds increase slightly, and the rise becomes more pronounced in 2015. The most significant jump occurs between 2015 and 2020, where the average maximum wind speed climbs to about 14 km/h, the highest value in the entire period. Overall, the pattern suggests that while wind speeds weakened initially, they have intensified in recent years, indicating a possible increase in strong wind events over time.

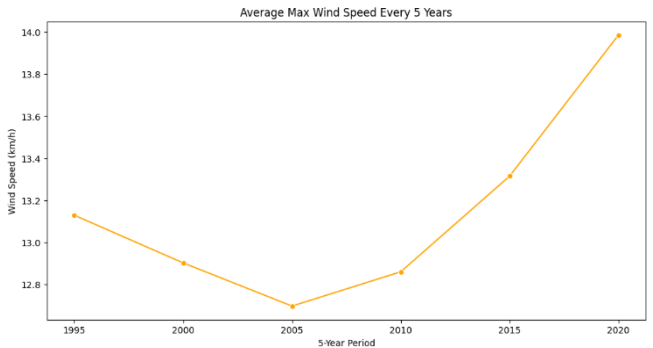


Fig 17: Windspeed Deviation

IV. DATA CORRELATIONS

1) Overall Correlation between Heat & Rainfall

i. Total Annual Rainfall

The chart shows how Bangladesh’s yearly rainfall changes over time. Some years receive significantly higher rainfall, while others show noticeable drops. This variation reflects natural shifts in monsoon strength and overall climate patterns across the decades.

ii. Average Monthly Rainfall

This chart displays how rainfall is distributed across the months of the year. Rainfall rises sharply during the monsoon months, peaks around mid-monsoon, and then gradually decreases. The dry months show very low rainfall, highlighting Bangladesh’s strong seasonal climate cycle.

iii. Average Monthly Humidity

The humidity chart shows that humidity increases during the rainy season and decreases in the dry months. The highest humidity coincides with peak monsoon rainfall, while the lowest levels appear in winter, when the weather is cooler and dry.

iv. Average Monthly Wind Speed

This chart illustrates how wind speed varies throughout the year. Certain months experience stronger winds, often aligned with pre-monsoon or monsoon activity, while other months remain relatively calm.

v. Rainfall Comparison Across Regions (5-Year)

The regional comparison chart highlights how different areas of Bangladesh receive varying amounts of rainfall over a five-year period. Some regions consistently get higher rainfall, while others remain comparatively lower, showing the influence of geography on climate.

vi. Monthly Rainfall vs. Humidity Correlation

This chart shows the relationship between rainfall and humidity. As rainfall increases, humidity generally rises as well, showing a positive correlation. This reflects how wetter conditions naturally lead to more moisture in the air.

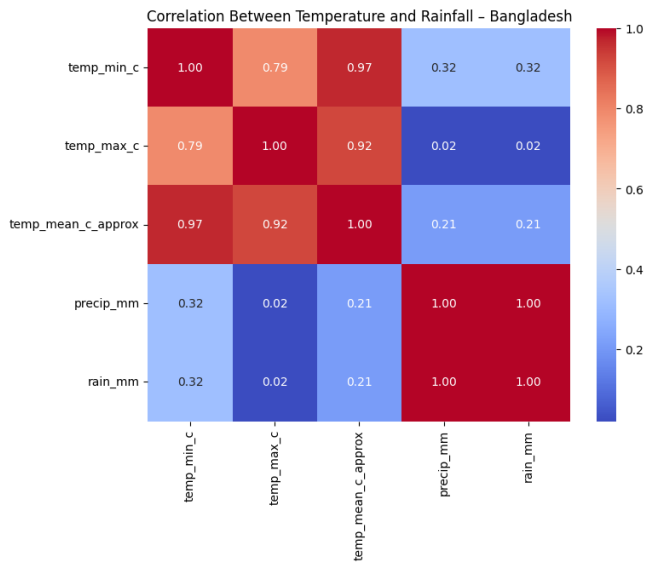


Fig 18: Correlation heatmap between temp & rainfall

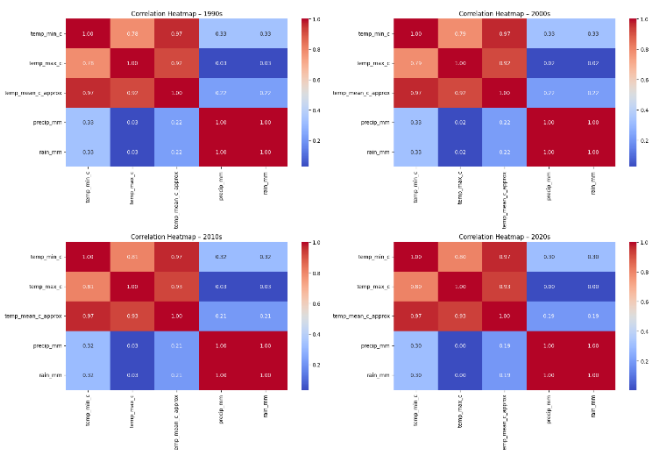


Fig 19: Correlation heatmap for every 5 years interval

2) Relationship between maximum temp & rainfall

This scatter plot illustrates how daily maximum temperature relates to the amount of rainfall. Most rainfall occurs when temperatures range from the mid-20s to around 32°C, where the concentration of points is highest. At lower and higher temperature extremes, rainfall appears much less frequent. The red trend line shows a very slight downward slope, indicating a weak negative relationship—meaning that as maximum temperature increases, rainfall tends to decrease slightly. However, the wide spread of points suggests that temperature alone is not a strong predictor of rainfall, and many other factors influence rainfall patterns.

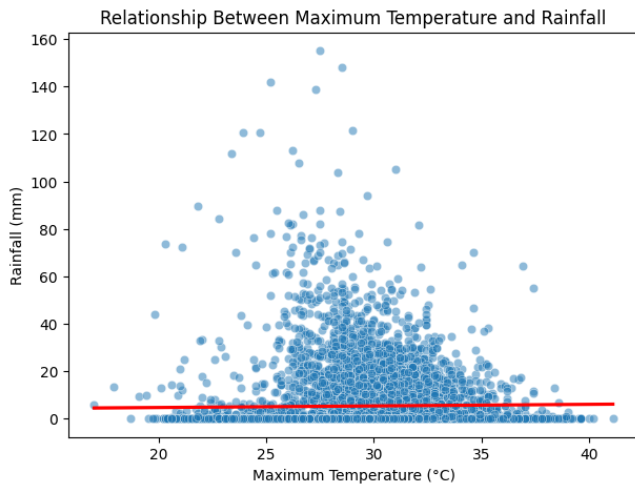


Fig 20: Relationship between maximum temp & rainfall

3) Predicted VS Actual Rainfall

This scatter plot compares predicted rainfall values with the actual recorded rainfall. Most of the data points cluster near the lower rainfall range, showing that the model frequently predicts small amounts of rainfall regardless of the actual value. The red diagonal line represents the ideal scenario where predictions perfectly match the observed rainfall. However, the predicted values rarely align with this line, especially for higher rainfall amounts, where the model consistently underestimates actual rainfall. Overall, the plot indicates that the model struggles to accurately capture higher rainfall events and tends to produce conservative predictions.

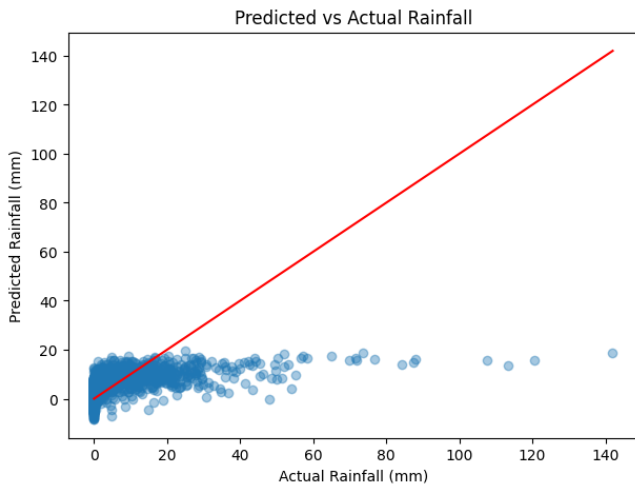


Fig 21: Predicted VS Actual Rainfall

4) Probability of rainfall after heatwave

This bar chart illustrates the likelihood of rainfall occurring after heatwave days compared to non-heatwave days. The probability of rainfall is significantly higher on days that are not preceded by a heatwave, reaching around 30%. In contrast, days following a heatwave show a much lower rainfall probability, close to 10%. This suggests that heatwave conditions may suppress immediate rainfall, likely due to atmospheric dryness and reduced moisture availability. Overall, the chart highlights a clear difference in rainfall patterns depending on whether or not a heatwave occurred the previous day.

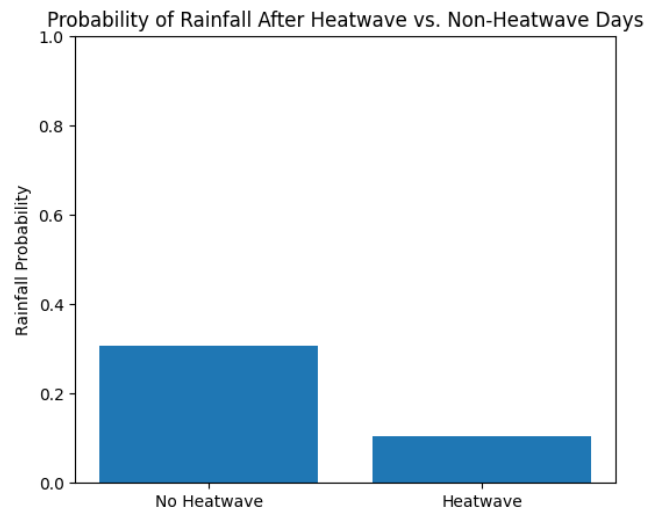


Fig 22: Probability of rainfall after heatwave

V. DATA ANALYSIS LOGISTICS

After thorough investigation and exploratory data analysis various results were found. We will try to deduce logistic precision of the outcomes and

A. Heatwave → Rainfall Logistic Model Summary

A logistic regression model was developed to quantify how heatwave conditions influence the probability of rainfall events. The model achieved an overall accuracy of **71%**, but this performance was dominated by correct predictions of the majority class (no-rain days). The classifier failed to identify rainfall events due to substantial class imbalance in the dataset.

Despite this limitation, the estimated regression coefficient for heatwave occurrence was informative:

$$\beta_1 = -1.291 \Rightarrow \text{Odds Ratio} = 0.27$$

This value indicates that the presence of a heatwave reduces the odds of rainfall by approximately **73%**, suggesting a strong inverse relationship between extreme heat and rain occurrence. This aligns with typical meteorological patterns, where heatwaves coincide with dry atmospheric conditions, suppressed convection, and reduced moisture availability. However, because the dataset contains far fewer rain events than non-rain days, the model defaulted to predicting the majority class. As a result, its predictive usefulness is limited, even though the estimated effect size remains meaningful. Improving the model would require addressing class imbalance (e.g., class weights, resampling techniques) or adding additional meteorological variables such as humidity, dew point, pressure, or wind patterns to capture the dynamics underlying rainfall formation.

This indicates that heatwaves decrease rainfall odds by approximately **73%**. However, model performance for predicting rainfall events was poor due to high class imbalance. Consequently, although the coefficient is meaningful, the classifier's utility is limited without resampling, class weighting, or additional meteorological predictors

B. Polynomial Regression Model (Degree 2): Performance Summary

A second-degree Polynomial Regression model was trained to evaluate the nonlinear relationship between heat variables and rainfall. The model achieved the following performance:

- Mean Absolute Error (MAE): 4.568
- Coefficient of Determination (R^2): 0.358

The MAE value indicates that, on average, the model's predictions deviate from actual rainfall values by approximately 4.57 mm, which is a moderate level of error for daily rainfall estimation. The R^2 score of 0.358 suggests that the model explains about 35.8% of the variance in rainfall, indicating a meaningful but incomplete capture of the underlying nonlinear patterns.

The polynomial transformation introduces interaction terms and squared components, enabling the model to represent curved relationships that a linear model cannot. This improved flexibility helps the regression detect trends between temperature indicators and rainfall intensity. However, the model still leaves a substantial amount of variability unexplained, implying that rainfall is influenced by additional meteorological factors not included in the current feature set (e.g., humidity, pressure, cloud cover, or large-scale weather systems).

These results show that while nonlinear modeling improves predictive ability over a purely linear approach, further enhancements—such as incorporating more atmospheric variables or using advanced models like Random Forests or Gradient Boosting—are likely needed for a more reliable rainfall estimation framework.

C. Error Handling

a) Error Distribution (KDE)

The KDE plot shows the smoothed distribution of prediction errors from the rainfall model. The curve is heavily skewed to the left, with most errors falling between -12 mm and 2 mm, indicating that the model frequently under-predicts rainfall. A long left tail stretches as far as -40 mm, meaning that on some days the model predicted significantly less rainfall than what actually occurred. The peak near slightly negative values suggests that small underestimations happen most often, while large overestimations are rare.

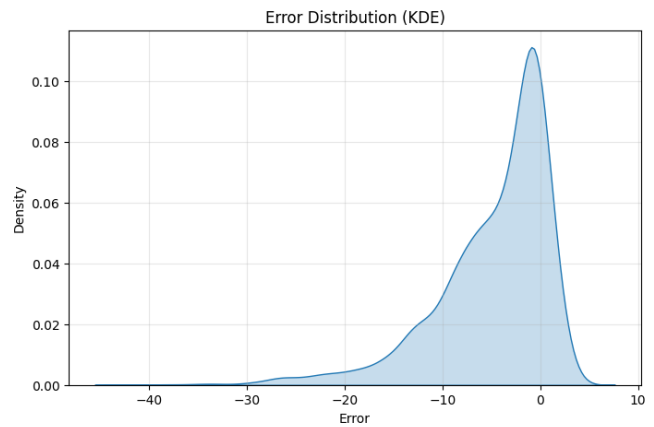


Fig 23: Error Distribution (KDE)

b) Residuals vs Predicted Values

This scatterplot compares model residuals (actual minus predicted) against predicted rainfall values. The downward diagonal pattern shows that as predicted rainfall increases, the residuals become increasingly negative. This pattern indicates a systematic bias in the model, where higher rainfall amounts are consistently under-predicted. A well-performing model would show randomly scattered points around zero, but here the strong linear trend reveals model miscalibration, especially for moderate to high rainfall predictions.

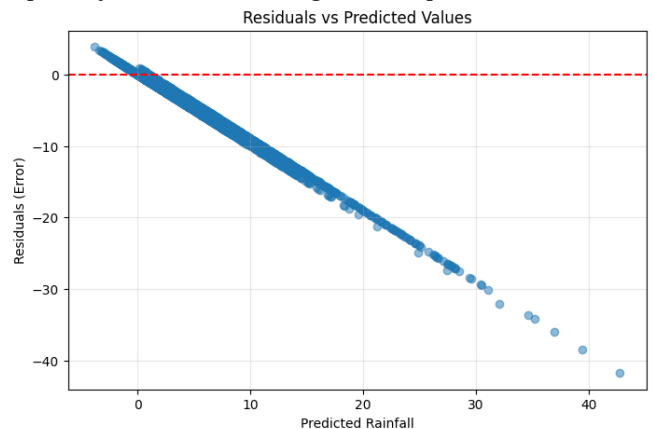


Fig 24: Residuals VS Predicted Values

c) Error Distribution (Histogram)

The histogram provides a frequency-based view of prediction errors. Most bars are concentrated around the range -10 to 0 mm, showing that the model typically predicts slightly less rainfall than what actually occurs. The histogram also has a long left tail extending toward -40 mm, similar to the KDE plot, confirming that large under-predictions, though less frequent, do occur. The near absence of bars on the positive side of the error axis indicates that over-predictions are uncommon, reinforcing the model's tendency to underestimate rainfall.

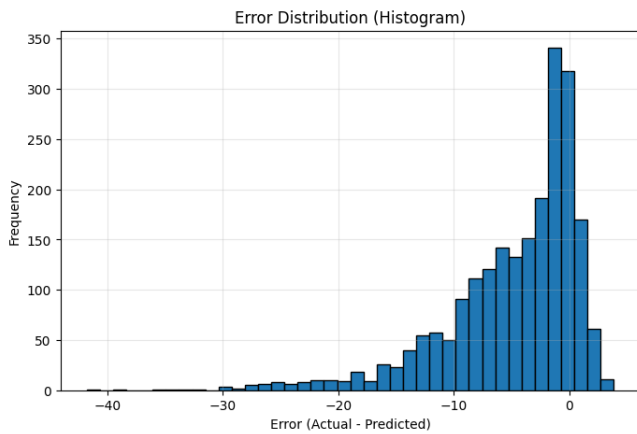


Fig 25: Error Distribution (Histogram)

d) Overall data integrity

The error analysis reveals a consistent pattern of underestimation by the rainfall prediction model. Both the Kernel Density Estimation (KDE) plot and the histogram show that most prediction errors fall between -12 mm and 2 mm, with errors concentrated around slightly negative values. This indicates that the model frequently predicts rainfall amounts that are slightly lower than the actual observed values. The long left tail extending toward -40 mm highlights that while less common, the model occasionally produces large under-predictions, especially during higher rainfall events. In contrast, large positive errors (over-predictions) are rare, reinforcing the model's overall downward bias.

The residuals vs. predicted values plot further confirms this behavior. Instead of being randomly scattered around zero—a sign of a well-calibrated model—the residuals form a strong downward-sloping line. As predicted rainfall increases, residuals become increasingly negative, meaning the model systematically underestimates higher rainfall values. This linear pattern indicates that the model's error grows progressively larger for heavier rain events, suggesting a significant lack of sensitivity to extreme rainfall. Overall, the combined evidence from the KDE distribution, histogram, and residuals scatterplot clearly shows that the model consistently predicts rainfall values that are lower than the observed amounts, with the bias becoming more pronounced as rainfall intensity increases.

VI. CONCLUSION

The overall analysis provides a comprehensive look into the relationship between temperature, rainfall, and extreme weather patterns in Bangladesh, supported by detailed visualizations and statistical exploration. The decade-wise correlation heatmaps consistently show strong internal relationships among temperature variables but reveal only weak to negligible correlations between temperature and rainfall. This suggests that rainfall patterns in Bangladesh are influenced by complex, non-linear factors rather than simple temperature fluctuations. Additional scatterplots reinforce this outcome, showing broad dispersion with no strong linear association between maximum temperature and rainfall.

The heatwave-focused analysis reveals that rainfall is less likely to occur on heatwave days compared to normal days, indicating that extreme heat conditions may temporarily suppress rainfall formation. However, this does not necessarily imply a long-term causal link—rather, it highlights short-term atmospheric interactions.

The machine learning modeling and error diagnostics further illustrate the challenge of predicting daily rainfall based solely on temperature-based inputs. The model consistently underestimates rainfall amounts, especially during higher rainfall events, as confirmed by the error distribution plots and residual patterns. The strong negative bias indicates that rainfall prediction requires additional meteorological variables beyond temperature—such as humidity, wind patterns, and atmospheric pressure—to achieve greater accuracy.

Taken together, the visual, statistical, and predictive analyses point to a clear conclusion: rainfall in Bangladesh exhibits highly variable and non-linear behavior that cannot be reliably explained or predicted using temperature alone. While temperature trends show consistent warming across decades, rainfall remains irregular, influenced by multiple atmospheric processes. This emphasizes the need for multi-factor models and deeper climatological data when assessing the impacts of climate change and forecasting rainfall in tropical regions like Bangladesh.

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All data processing, visualization, and interpretation were carried out independently, ensuring full control over each step and maintaining the clarity, accuracy, and reliability of the study.

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Link for code:

<https://www.kaggle.com/code/kazisadathossain/stat-final-thesis?scriptId=285198540>

<https://www.kaggle.com/code/kazisadathossain/statvisual?scriptId=285198923>